

Noormah Rizwan

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Github: [Link to Code](#)

EDUCATION

Michigan State University

Ph.D. Agricultural, Food & Resource Economics

Specialization: Environmental and Resource Economics

Dissertation Title: Adaptation to Natural and Artificial Water Scarcity

East Lansing, MI

2021 - Expected August 2026

Georgetown University

M.A. Applied Economics

Washington, DC

2016 – 2017

Lahore University of Management Sciences

B.Sc. (Hons) Economics

Lahore, Pakistan

2011 – 2015

RESEARCH

Research Interests: Applied Microeconomics; Environmental and Climate Economics; Development Economics; Urban and Agricultural Resource Governance; Structural and Reduced-Form Microeconometrics; Spatial Data and Geographical Analysis; Behavioral Economics

Papers Under Review

Noormah Rizwan, Jamshid Jalali, Molly Sears, Sean A Woznicki, Tao Liu, Oskar Marko, Mirjana Radulović. “Irrigation in a Changing Landscape: A Combined Economic and Hydrological Approach.” (*Revise and Resubmit, Water Resources Research*). ([Link](#))

Abstract: Warmer temperatures and erratic rainfall threaten global water availability. While irrigation is a key risk-mitigation strategy, limited research has examined how farmers’ incentives to irrigate evolve under climate change, or the effects of large-scale irrigation expansion on future water availability. We take an integrated econometric–hydrologic modeling approach to study how short- and long-term climate fluctuations influence irrigation incentives and their implications for future water resources. Using discrete choice model, we identify when and where irrigation becomes a viable option under climate change and then integrate the results irrigation-enabled fields into the Soil and Water Assessment Tool Plus (SWAT+) to determine realized irrigation activity, irrigation water use, and the implications for future water availability. We find that farmers respond to both short- and long-term climate signals, with warmer and drier conditions strengthening incentives for irrigation. At the same time, widespread irrigation expansion deepens regional water deficits, creating a tension between individual resilience and aggregate sustainability.

Working Papers

Noormah Rizwan, Molly Sears, Sean A Woznicki, Oskar Marko. “Weathering the Change: Modeling Crop Choices in Response to Climate Variability.” (*Job Market Paper - To be Presented at Job Seminar*) ([Link](#))

Abstract: Agricultural land allocation remains sensitive to climate change. As temperatures rise and precipitation patterns become more variable, farmers switch to crops better suited to the changing climate. While this substitution has tremendous market and welfare implications, quantifying these impacts remains understudied, largely due to limited data on adaptive decisions and challenges in measuring profitability. Using field-level cropping data, we examine how climate change shapes crop choices. First, we estimate a yield–weather model to identify crop-specific thresholds for extreme events and project climate impacts on yields. We then integrate these results into a discrete crop choice model to assess how shifts in yields and profitability influence farmer decisions. We run three simulations: (i) climate change alone, (ii) with yield effects, and (iii) with yield and market effects,

where prices adjust to yield shifts. We find that under changing climatic and market conditions; farmers switch away from profitable crops such as maize and soybean to less profitable but more resilient wheat and sunflower. These results have implications for farmers' welfare as wheat and sunflower are 1.5x less profitable compared to maize and soybean.

Noormah Rizwan, James Sears. "Can the Water Mafia Be Good? Evidence from Karachi's Residential Water Market."

Abstract: In this paper, we investigate how households and non-state actors adapt to chronic water scarcity and infrastructural failures in developing countries. In many urban areas, unreliable and inadequate public water supply have given rise to the "water mafia" — non-state actors that extract and sell water, often at high prices, to households disconnected from the piped network. The overall social welfare impact of these actors remains uncertain due to limited microdata on household-level adaptation to piped water scarcity. Leveraging a novel household survey that I designed and implemented in Karachi, Pakistan, I estimate a discrete-continuous model of household water demand and quantify the welfare changes from a counterfactual eliminating the water mafia. Preliminary findings reveal substantial disparities in piped water access, forcing poorer households to rely on illegal water vendors, creating opportunities for the mafia to engage in exploitative behavior.

Policy Publications

Noormah Rizwan, Sania Haider Shikoh, Stephen Davies, Muhammad Saad Moeen, Abdul Wajid Rana, Zeeshan Haider. "Assessing the Economic Cost of Depleting Groundwater in Balochistan: A SAM Multiplier Approach." (*IFPRI Discussion Paper*, 2021). ([Link](#))

Muhammad Saad Moeen, Zeeshan Haider, Sania Haider Shikoh, **Noormah Rizwan**, Amna Ejaz, Stephen Davies, Abdul Wajid Rana. "Estimating the Economic Impacts of the First Wave of COVID-19 in Pakistan using a SAM Multiplier Model." (*IFPRI Discussion Paper*, 2021). ([Link](#))

Abdul Wajid Rana, Stephen Davies, Muhammad Saad Moeen, Sania Haider Shikoh, **Noormah Rizwan**. "Solarization of Electric Tube-wells for Agriculture in Balochistan: Economic and Environmental Viability." (*IFPRI PACE Working Paper*, 2020). ([Link](#))

Daniel Jeongdae Lee, **Noormah Rizwan**. "What Explains Divergent Investment Performances in Asia-Pacific?" (*MPFD Policy Brief*, 2018). ([Link](#))

PROFESSIONAL EXPERIENCE

Research Assistant Aug 2021 – Present

Michigan State University, East Lansing, MI

Working on a multi-disciplinary project investigating climate adaptation in agriculture, focusing on crop choices and irrigation adoption in Serbia.

Research Analyst Jul 2019 – Jul 2021

International Food Policy Research Institute, Islamabad, Pakistan

Developed policy-oriented research on agricultural resilience and resource management, providing analysis to support federal and provincial decision-making.

Program Assistant Oct 2018 – Jan 2019

Punjab Commission on the Status of Women, Lahore, Pakistan

Cleaned and analyzed survey data on women's economic well-being in Punjab, contributing to policy briefs for provincial government.

Intern Oct 2017 – Jan 2018

UN ESCAP, Macroeconomic and Finance Division, Bangkok, Thailand

Conducted quantitative analysis of investment performance in Asia-Pacific, contributing to a UN policy brief.

MENTORING & TRAINING

Designed and led a week-long training program for undergraduate enumerators in Karachi, covering survey implementation, field protocols, and adaptive interviewing techniques.

Conducted a two-day workshop on data entry and cleaning in Stata, training undergraduate enumerators to produce analysis-ready datasets and apply basic econometric techniques.

POLICY ENGAGEMENT & MEDIA OUTREACH

Informed federal decision-making in Pakistan by presenting groundwater research to the Ministry of Energy, contributing to the rejection of a proposed solar tubewell subsidy due to its implications for groundwater depletion.

Presented technical findings from *Valuation of Water Flows into the Indus Delta* to policymakers; drew lessons on effective translation of economic research for policy audiences.

Featured in media: *Who pays for water? MSU researchers examine Karachi's water mafia, local experts reflect on Michigan's water woes* ([Link](#))

CONFERENCE PRESENTATIONS

Universities Council on Water Resources (UCOWR) Conference	July 2025
Agricultural and Applied Economics Association Meeting	July 2025
Association of Environmental and Resource Economists	May 2025
Fate of the Earth, MSU Water Alliance	March 2025
Heartland Environmental and Resource Economics Workshop, UIUC	October 2024
MSU & University of Michigan Energy Environmental Econ Day	August 2024
IFPRI Retreat and Research (RISE) Conference	November 2020

RESEARCH GRANTS

American Institute of Pakistan Studies Short-term Research Grant	2024
Karachi Urban Water Market Survey (\$3000)	
A. Allan Schmid Fellowship in Institutional Economics (Michigan State University)	2024
Karachi Urban Water Market Survey (\$8700)	

SCHOLARSHIPS & AWARDS

Fulbright Scholar	2016-2017
Georgetown University	
Graduation with Distinction	2015
Lahore University of Management Sciences	

ACADEMIC SERVICE

Peer Reviewer, Water Resources Research (AGU)

Provided formal review for “*Water Scarcity Compounds the Negative Effects of Salinity on Irrigated Agriculture.*”

Recruitment Speaker

Presented research as part of a departmental recruitment event at UMich School for Environment and Sustainability (SEAS).

Graduate Student Orientation Panelist

Advised incoming graduate students on program expectations and resource navigation.

TECHNICAL SKILLS AND LANGUAGES

Software: Stata; R; GIS; LaTeX; GAMS

Languages: Urdu, English

REFERENCES

Dr. Molly Sears

Assistant Professor
Department of Agricultural, Food, and Resource
Economics
Michigan State University
Email: searsmol@msu.edu

Dr. Soren Anderson

Associate Professor
Department of Agricultural, Food, and Resource
Economics
Michigan State University
Email: sta@msu.edu

Dr. James Sears

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Department of Agricultural, Food, and Resource
Economics
Michigan State University
Email: searsja1@msu.edu

Dr. Mywish Maredia

Professor
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Economics
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